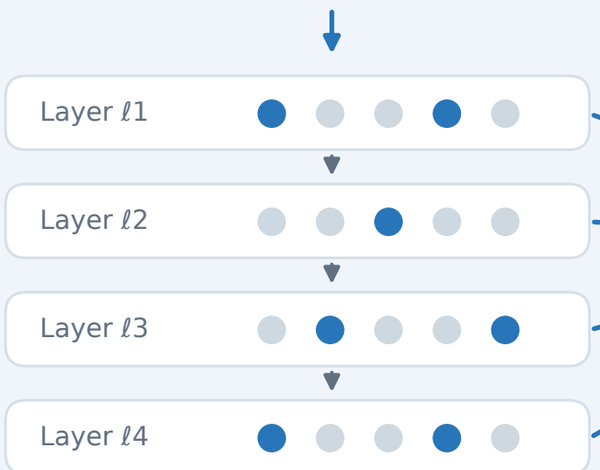
B / Two model runs, connected by one activation vector

Source head outputs are pooled across layers; the pooled direction is added at multiple probe layers.

(a) SOURCE RUN

Keep palindromes starting with a vowel.

```
ys = [s for s in xs
      if s == s[::-1]]
zs =
```



Read selected heads at one token t

Extraction layers determined by H

PROJECT & SUM

$$g_t = \sum_{(\ell,h)\in H} W_O^{\ell,h} a_t^{\ell,h}$$

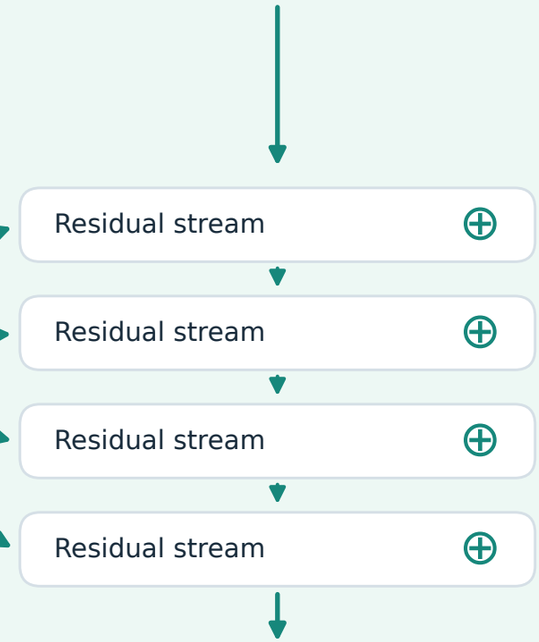
Normalize to $\hat{g}_t$

Arrows carry projected head outputs, not text or source residual states.

(b) PROBE RUN · SAME MODEL

“What are you doing at this line …?”

Fixed prompt; no source code text



“Keep vowel-initial strings.”

Depth ↓ · Layers shown schematically

$$r_{\ell,s} \leftarrow r_{\ell,s} + \alpha \|r_{\ell,s}\| \hat{g}_t$$

SCHEMATIC EXAMPLES · Verbalizations are illustrative, not recorded outputs. | Additive norm-relative GoalScope.